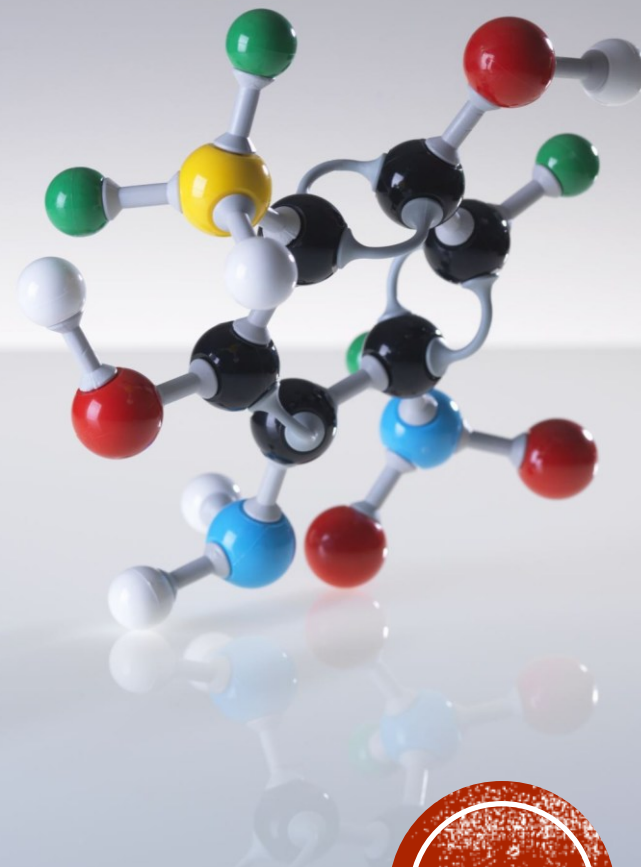


MOLECULAR, DOCKING VISUALIZATION AND REPRESENTATIONS

**Dr. Oluyemi Wande | Dr. Adeniyi Adewumi |
Shadrach Eze**



1

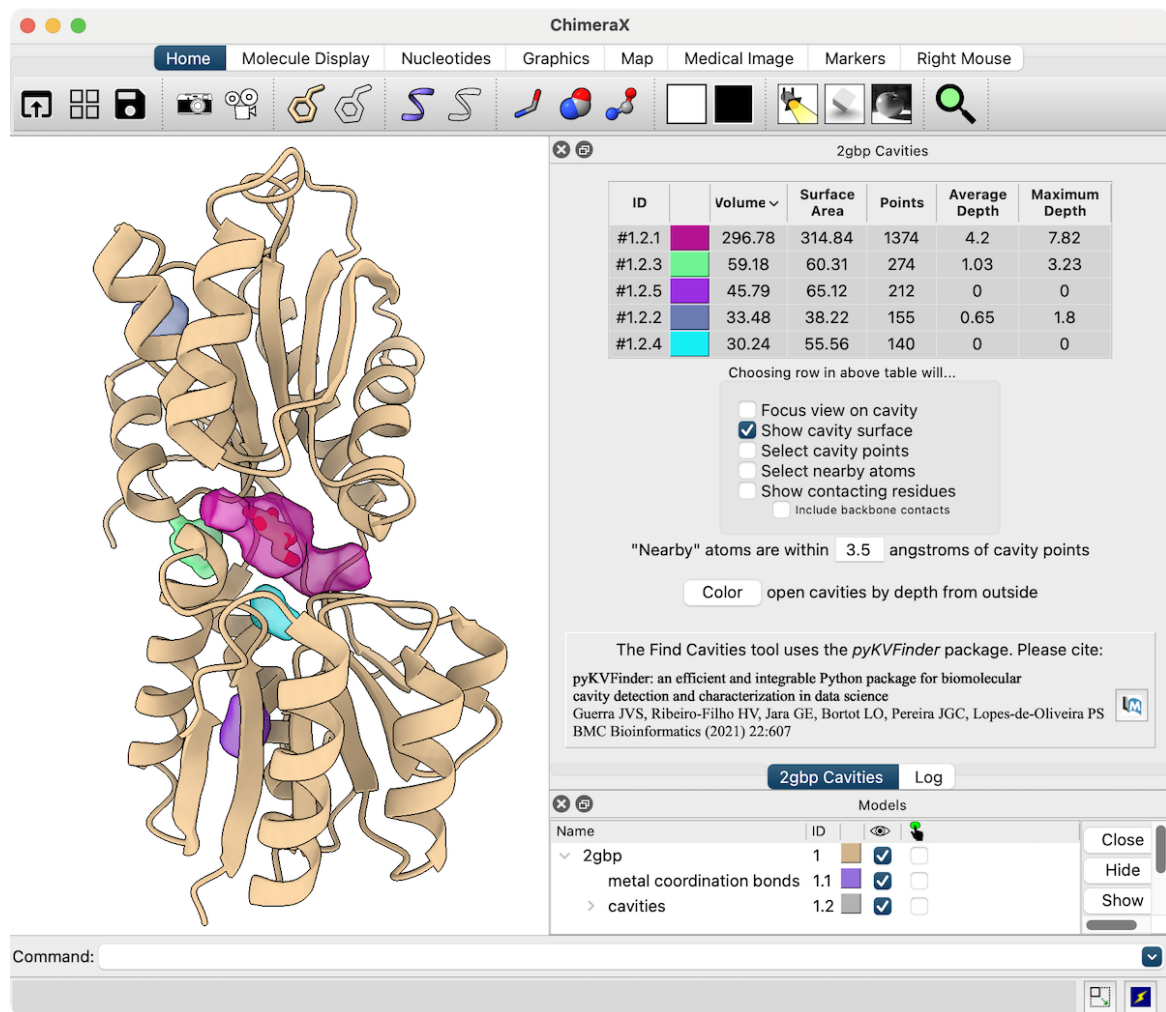
CONTENTS

Chimera

Discovery
Studio

Pymol

ChimeraX



ChimeraX

Home Molecule Display Nucleotides Graphics Map Medical Image Markers Right Mouse

2gbp Cavities

ID	Volume	Surface Area	Points	Average Depth	Maximum Depth
#1.2.1	296.78	314.84	1374	4.2	7.82
#1.2.3	59.18	60.31	274	1.03	3.23
#1.2.5	45.79	65.12	212	0	0
#1.2.2	33.48	38.22	155	0.65	1.8
#1.2.4	30.24	55.56	140	0	0

Choosing row in above table will...

- ☐ Focus view on cavity
- ☒ Show cavity surface
- ☐ Select cavity points
- ☐ Select nearby atoms
- ☐ Show contacting residues
- ☐ Include backbone contacts

"Nearby" atoms are within 3.5 angstroms of cavity points

Color open cavities by depth from outside

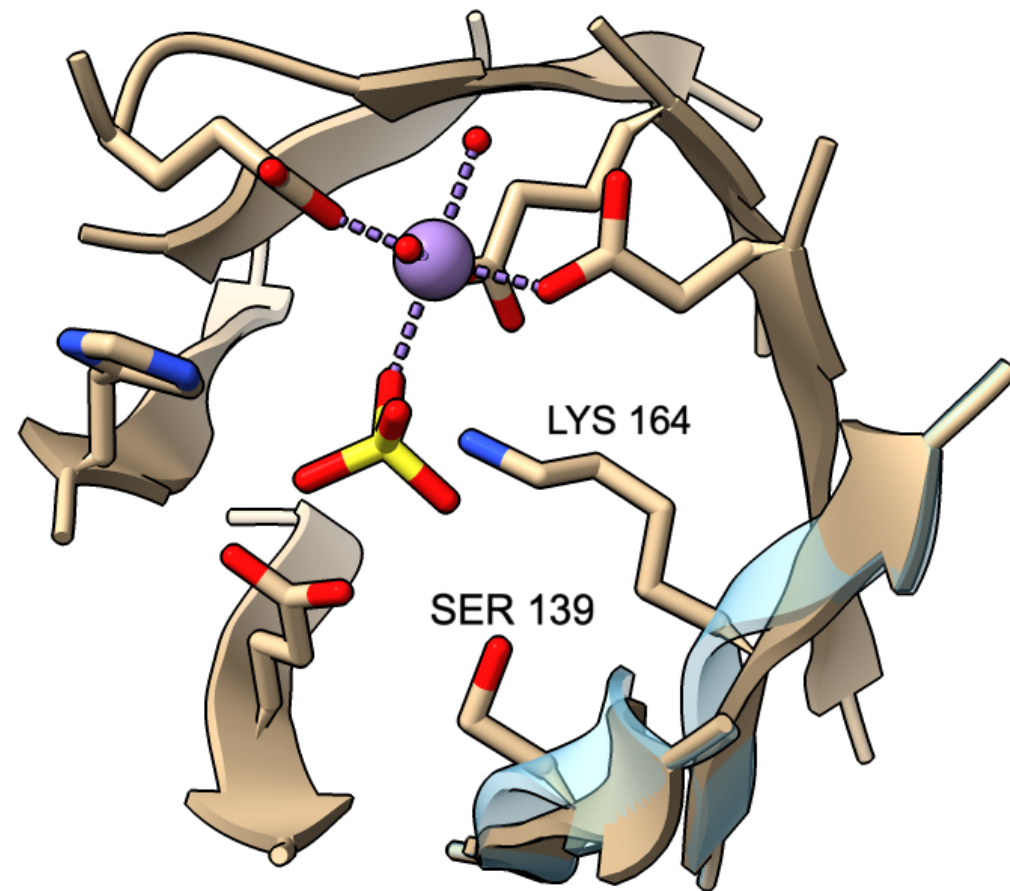
The Find Cavities tool uses the *pyKVfinder* package. Please cite:
pyKVfinder: an efficient and integrable Python package for biomolecular cavity detection and characterization in data science
 Guerra JVS, Ribeiro-Filho HV, Jara GE, Bortot LO, Pereira JGC, Lopes-de-Oliveira PS
 BMC Bioinformatics (2021) 22:607

2gbp Cavities Log

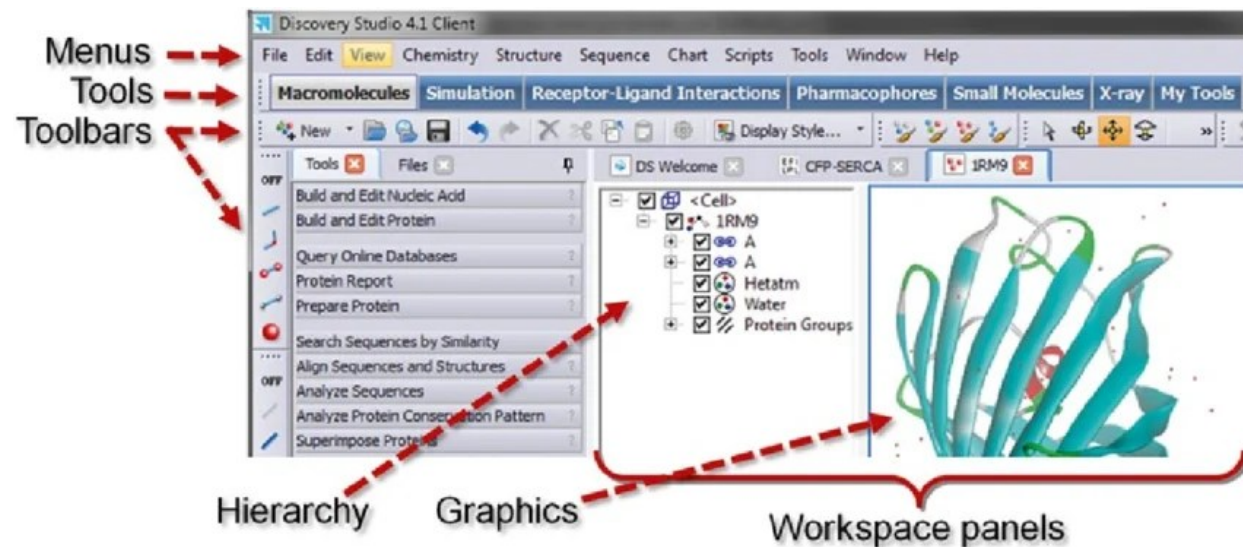
Models

Name	ID	View	Hide	Show
2gbp	1	☒	☐	☐
metal coordination bonds	1.1	☒	☐	☐
cavities	1.2	☒	☐	☐

Command:



Discovery Studio



&

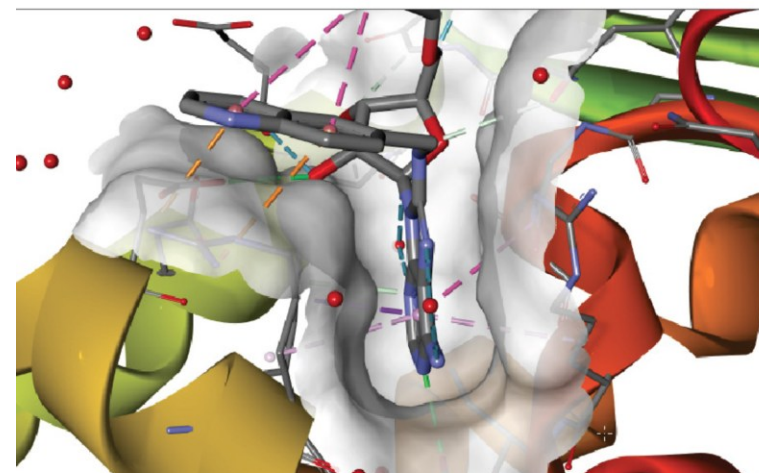
Pharmacophore Design

Structure Based Design

QSAR/ADMET Modelling



**BIOVIA DISCOVERY STUDIO®
VISUALIZER**
DATASHEET

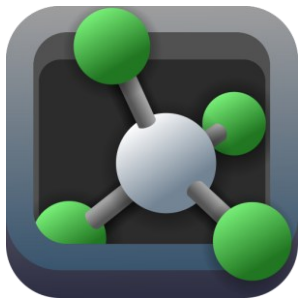


**FREE
VISUALIZATION
AND ANALYSIS
SUITE**

If you need access to commercial-grade 3D molecular visualization tools for viewing, sharing and analyzing protein and small molecule data, then BIOVIA Discovery Studio Visualizer (DS Visualizer) offers an extensive set of free functionality.

BIOVIA DS Visualizer is a free, feature-rich molecular modeling environment, for both small molecule and macromolecule applications. Built from, and fully compatible with the latest release of our enterprise-ready expert modeling product, BIOVIA Discovery Studio 4.0, experts and colleagues can seamlessly and efficiently exchange results, without loss of either time or scientific information.

Molecular Dynamics



PyMol

Installation



Windows
EXE Installer

Windows
ZIP Archive

macOS
DMG Disk Image

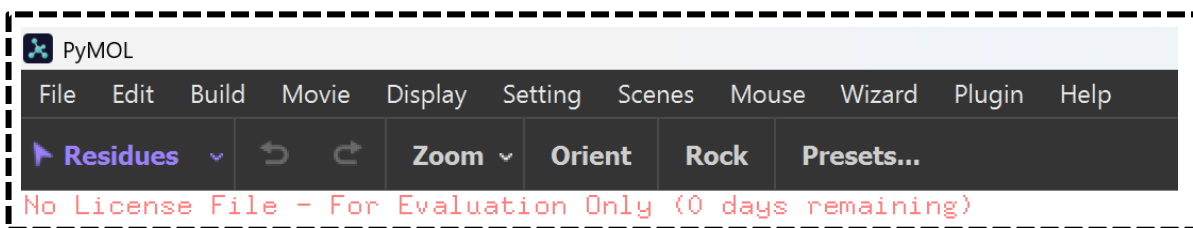
Linux
TAR.BZ2 Archive

Or install from the Schrodinger Anaconda Channel.

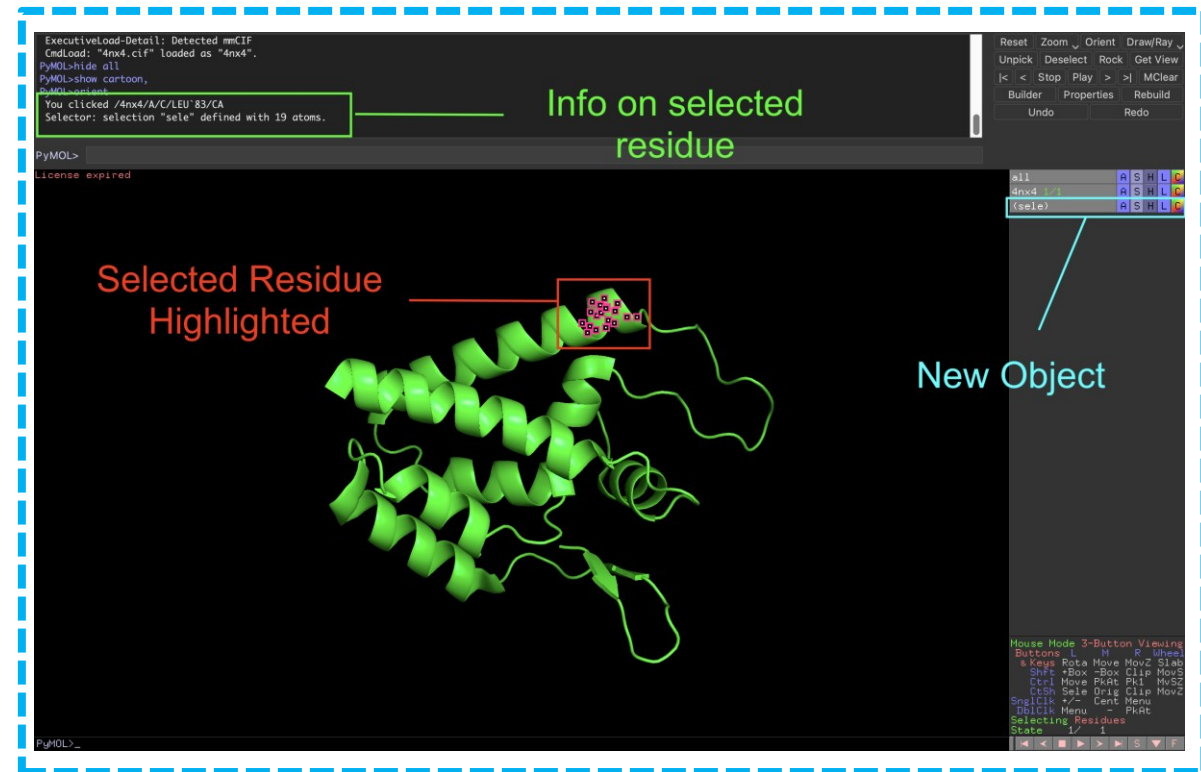
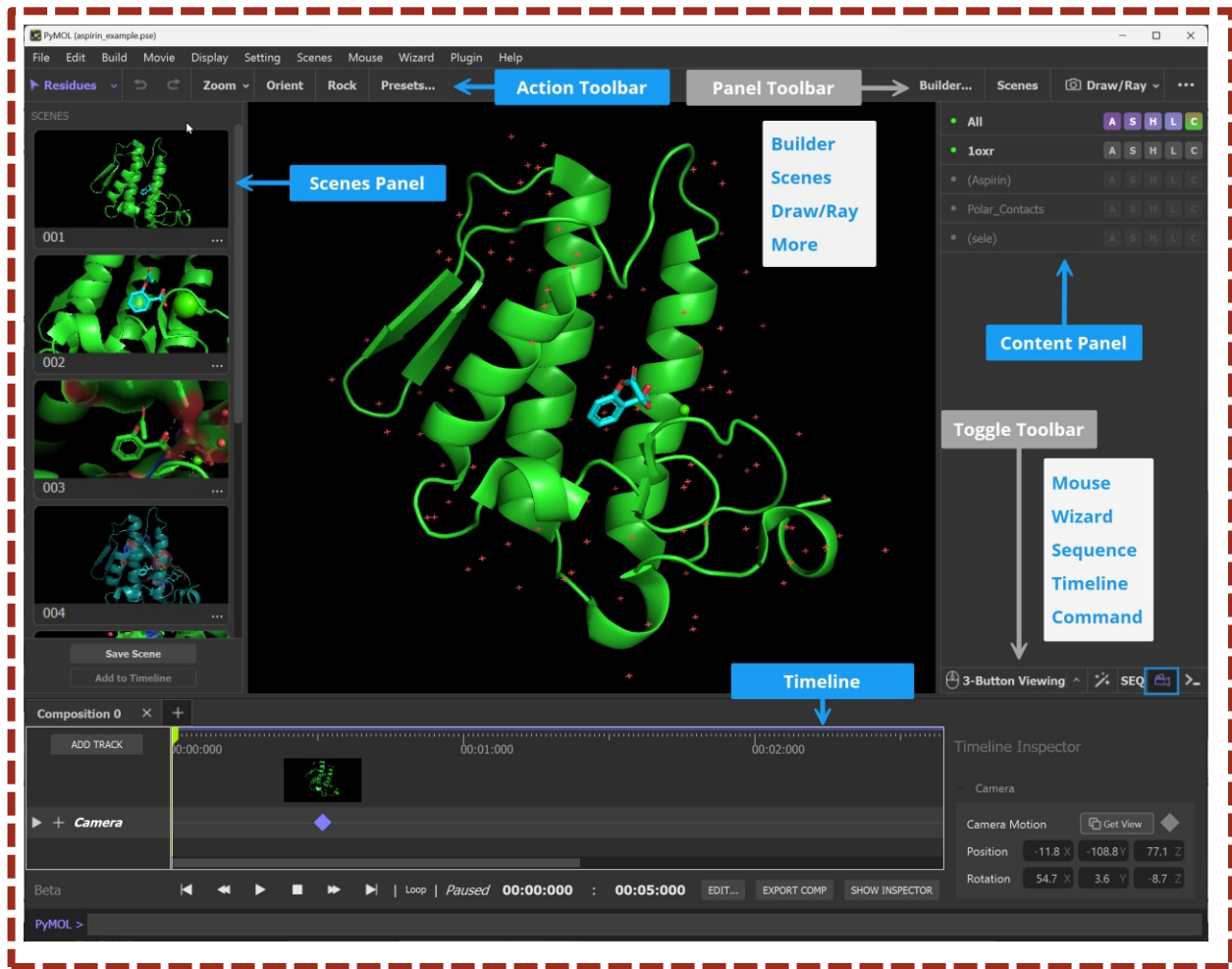
```
conda install -c conda-forge -c schrodinger pymol-bundle
```

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Existing users: [DOWNLOAD LICENSE FILE](#)



- **Needs a license from Schrodinger**
- **However, can be installed without a license**



Have a robust **command line** which wraps **python code** for **easy of scripting**

EXTRA RESOURCES

- <https://diegomariano.com/tutorial-chimerax/>
- <https://zenodo.org/records/8154327>
- https://www.adrianomartinelli.it/Fondamenti/Tutorial_0.pdf
- <https://dasher.wustl.edu/bio5357/software/pymol/simple-tutorial.pdf>